CEBT II REACTION ENGINEERING

2. Reactions of solids

- In this section we will consider non-catalytic reactions in which a solid is one of the reagents.
- Specifically we will consider overall kinetics for reactions between a solid and a fluid.
- In some cases the solid reacts by the action of heat alone, without needing another reagent.
  - In that case, heat transport is the critical element affecting the kinetics.
  - This is often the case with cooking
- In many cases a reactant must diffuse from the fluid (in)to the solid. The solid may either shrink as it reactor or not (shrinking particle vs shrinking core model).

2.1 Shrinking particle model

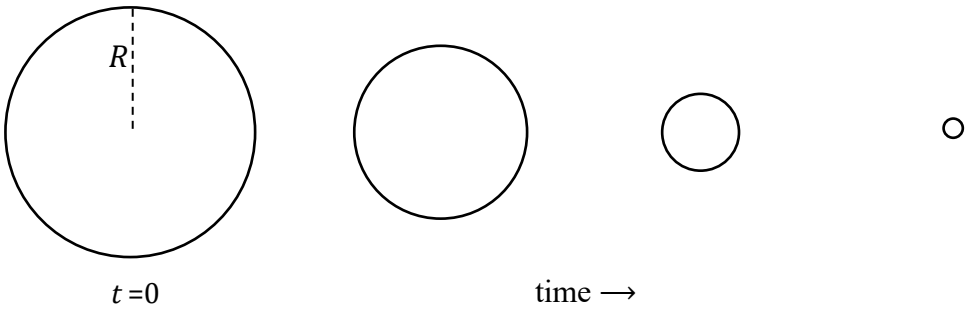
- Consider that a solid reacts with a gas (or liquid) to give a gaseous (or liquid) product. The overall reaction is

  $\nu_A\ A_{(s)} + \nu_B\ B_{(g)} \longrightarrow \text{products}_{(g)}$
- Examples are

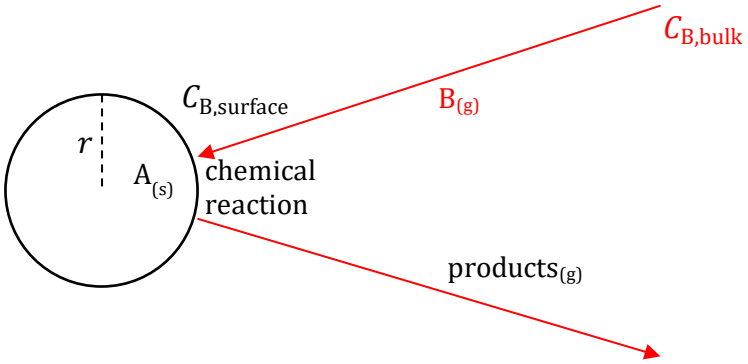
  $C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)}$

  $C_{(s)} + H_2O_{(g)} \longrightarrow CO_{(g)} + H_{2(g)}$
- The solid will decrease in size during the reaction.
- For simplicity, we shall assume that
  - each particle of solid A is spherical; the initial radius will be R, while the radius at a particular moment in time will be r.
  - each particle of solid A is non-porous.
  - B is a gas (though treatment of a liquid is analogous).

Shrinking particle model



At a particular moment:



- The overall rate of reaction may be affected by
  - external mass transport of gas B to the surface of the particle (or of products away from the particle).
  - the intrinsic kinetics of the chemical reaction
- We shall consider in turn what is the overall rate of reaction when each of these is the rate-limiting step.
  - We calculate the instantaneous rate of reaction for a particular particle size – this is a pseudo-steady state calculation that assumes radius is constant. We then use that result to calculate how the radius of the particle changes with time.

2.1.1 Shrinking particle with external mass transport as the slowest step

- Consider the overall reaction

  $\nu_A\ A_{(s)} + \nu_B\ B_{(g)} \longrightarrow \text{products}_{(g)}$

with external mass transport being the rate determining step.

- Let us use a mass transport coefficient k_{g} for transport in the gas phase.
  - An alternative approach would be to use the diffusion equation (with or without the convection term) to describe mass transport in the gas phase.
- Moles per second of gas B travelling to surface

  $= 4\pi r^2 k_{\mathrm{g}}\left(C_{\mathrm{B,bulk}} - C_{\mathrm{B,surface}}\right)$
- In this case the intrinsic rate of the chemical reaction is relatively fast. This implies

  $C_{\mathrm{B,surface}} \approx 0$
- Moles per second of solid A reacting

  $= 4\pi r^2 k_{\mathrm{g}} C_{\mathrm{B,bulk}} \dfrac{\nu_{\mathrm{A}}}{\nu_{\mathrm{B}}}$
- But moles per second of solid A reacting

  $= \dfrac{-d}{dt}\left(\dfrac{4}{3}\pi r^3 \rho_{\mathrm{m}}\right)$

  where ρ_{m} is the molar density of the solid
- Hence

  $$-4\pi r^2 \rho_{\mathrm{m}} \frac{dr}{dt} = 4\pi r^2 k_{\mathrm{g}} C_{\mathrm{B,bulk}} \frac{\nu_{\mathrm{A}}}{\nu_{\mathrm{B}}}$$

  $$\frac{dr}{dt} = -\frac{k_{\mathrm{g}} C_{\mathrm{B,bulk}}}{\rho_{\mathrm{m}}} \frac{\nu_{\mathrm{A}}}{\nu_{\mathrm{B}}}$$
- The mass transport coefficient k_g usually depends on radius. It is normally determined from a correlation for Sherwood number, e.g. $Sh = 2 + 0.6(Sc)^{1/3}(Re)^{1/2}$ [valid for mass transfer to or from a sphere for Re < 200 and Sc < 250].

Special case

- In the Stokes law regime (no gas flow or very small particles),

  $$Sh = 2 = \frac{k_{\mathrm{g}} d}{D_{\mathrm{bulk}}}$$

  $$\frac{dr}{dt} = -\frac{2D_{\mathrm{bulk}}}{2r} \frac{C_{\mathrm{B,bulk}}}{\rho_{\mathrm{m}}} \frac{\nu_{\mathrm{A}}}{\nu_{\mathrm{B}}}$$
- Integrate with boundary condition that $r = R$ at time $t = 0$

  $$\int_R^r -r dr = \int_0^t \frac{D_{\mathrm{bulk}} C_{\mathrm{B,bulk}}}{\rho_{\mathrm{m}}} \frac{\nu_{\mathrm{A}}}{\nu_{\mathrm{B}}}\, dt$$

  $$\frac{1}{2}(R^2 - r^2) = \frac{D_{\mathrm{bulk}} C_{\mathrm{B,bulk}}}{\rho_{\mathrm{m}}} \frac{\nu_{\mathrm{A}}}{\nu_{\mathrm{B}}} t$$

- The time for complete reaction, τ_c, is the time taken to reach $r = 0$.
  - τ_c is called the burn-out time if it is a combustion reaction.

$$\tau_c = \frac{R^2 \rho_m}{2 D_{bulk} C_{B,bulk}} \frac{\nu_B}{\nu_A}$$

- The particle radius varies with time according to

$$\frac{r}{R} = \sqrt{1 - \frac{2 D_{bulk} C_{B,bulk}}{\rho_m R^2} \frac{\nu_A}{\nu_B} t}$$

$$\frac{r}{R} = \sqrt{1 - \frac{t}{\tau_c}}$$

- The fractional conversion of the particle is

$$X = 1 - \frac{\frac{4}{3}\pi r^3 \rho_m}{\frac{4}{3}\pi R^3 \rho_m} = 1 - \left(\frac{r}{R}\right)^3$$

$$X = 1 - \left(1 - \frac{t}{\tau_c}\right)^{3/2}$$

2.1.2 Shrinking particle with chemical reaction as the slowest step

- Consider the overall reaction

$$\nu_A \, A_{(s)} + \nu_B \, B_{(g)} \longrightarrow \text{products}_{(g)}$$

- For this example, we shall assume that the reaction is first-order with respect to gas B with rate constant, per unit surface area of solid, of k_{SA}.
- Moles per second of B reacting

$$= \nu_B 4\pi r^2 k_{SA} C_{B,surface}$$

- If external mass transport is relatively rapid, then

$$C_{B,surface} \approx C_{B,bulk}$$

- Moles per second of solid A reacting

$$= \nu_A 4\pi r^2 k_{SA} C_{B,bulk}$$

- But moles per second of solid A reacting

$$= -\frac{d}{dt}\left(\frac{4}{3}\pi r^3 \rho_m\right)$$

where ρ_m is the molar density of the solid

- Hence

$$-4\pi r^2 \rho_m \frac{dr}{dt} = \nu_A 4\pi r^2 k_{SA} C_{B,bulk}$$

$$\frac{dr}{dt} = -\nu_A \frac{k_{SA} C_{B,bulk}}{\rho_m}$$

- Integrate with boundary condition that $r = R$ at time $t = 0$

$$\int_R^r -dr = \int_0^t \nu_A \frac{k_{SA} C_{B,bulk}}{\rho_m} dt$$

- If k_{SA} is constant with time (which is okay if temperature is not changing)

$$R - r = \nu_A \frac{k_{SA} C_{B,bulk}}{\rho_m} t$$

- The time for complete reaction (when r becomes zero) will be

$$\tau_c = \frac{1}{\nu_A} \frac{\rho_m}{k_{SA} C_{B,bulk}} R$$

- The particle radius therefore varies with time according to

$$r = R - \nu_A \frac{k_{SA} C_{B,bulk}}{\rho_m} t$$

$$\frac{r}{R} = 1 - \frac{t}{\tau_c}$$

- The fractional conversion of the particle

$$X = 1 - \frac{\frac{4}{3}\pi r^3 \rho_m}{\frac{4}{3}\pi R^3 \rho_m} = 1 - \left(\frac{r}{R}\right)^3$$

$$X = 1 - \left(1 - \frac{t}{\tau_c}\right)^3$$

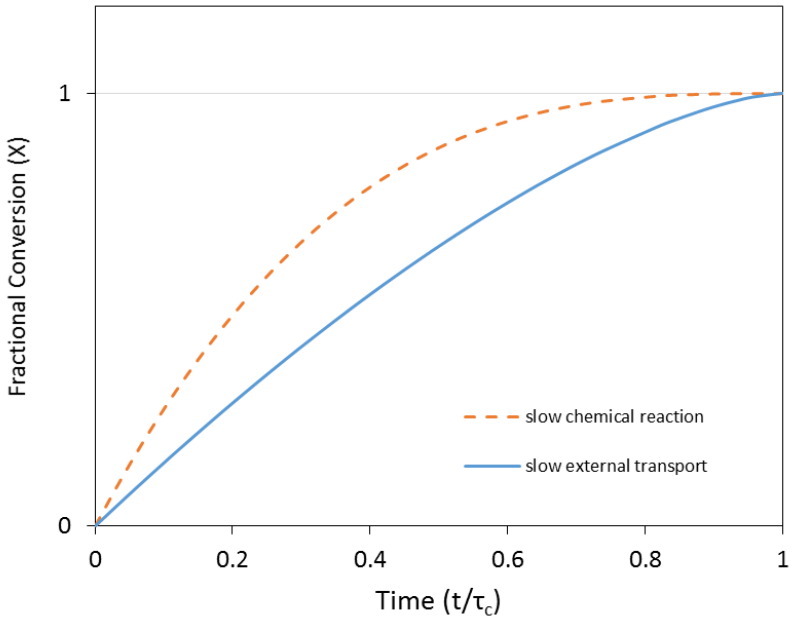


Figure: Fractional conversion for the limiting cases of the shrinking particle model

2.2 Shrinking core model

- Consider that a solid reacts with a gas (or liquid) to give a solid product. The overall reaction is

  $\nu_A\ A_{(s)} + \nu_B\ B_{(g)} \longrightarrow \nu_C\ C_{(s)} + \text{products}_{(g)}$

- Commercially important examples of this case are

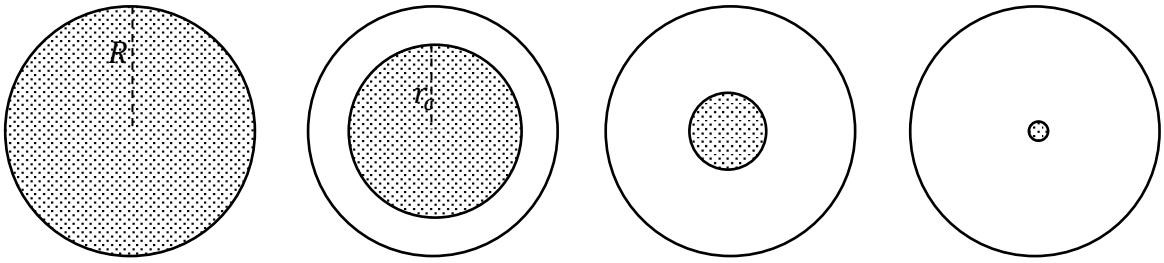
  $ZnS_{(s)} + O_{2(g)} \longrightarrow ZnO_{(s)} + SO_{2(g)}$

  $Fe_3O_{4(s)} + 4H_{2(g)} \longrightarrow 3Fe_{(s)} + 4H_2O_{(g)}$

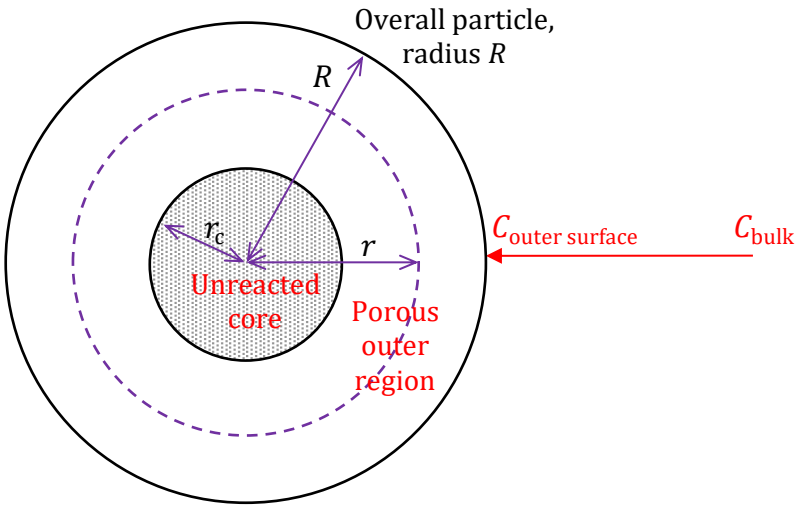
  Catalyst with coke in pores + O_2 ⟶ Regenerated porous catalyst + $CO_{2(g)}$

- The reaction will stop if product C is non-porous because a passivation layer around solid A would prevent further reaction (see CEBT II Corrosion).
- However, the reaction will continue if solid C is porous to the gas.
- For simplicity, we shall assume that
  - the particle of solid A is spherical and non-porous
  - the overall particle does not change size significantly.

Shrinking core model



At a particular moment:



2.2.1 Shrinking core with internal diffusion as the slowest step

- In this case there will be a concentration profile of incoming gaseous reagent in the porous outer region of the particle.
- Let us use an effective diffusion coefficient D_e for the gas in the outer porous region.
  - We will discuss diffusion in porous solids more in Section 6.
- Moles per second of gas B travelling *inwards* at a radius r

  $$= D_e \left(\frac{dC}{dr}\right) 4\pi r^2$$

- Because the gas does not react in this region, this must be independent of r (for the region $r_c < r < R$).

  $$\frac{d}{dr}\left(D_e \frac{dC}{dr} 4\pi r^2\right) = 0$$

- Hence

  $$r^2 \frac{dC}{dr} = \text{constant}$$

  $$C = \frac{a}{r} + b$$

- The boundary conditions to find a and b are
  - $C \approx C_{\text{bulk}}$ at $r = R$ if external mass transport is significantly faster than the internal mass transport.
  - $C \approx 0$ at $r = r_c$ if chemical reaction is significantly faster than the internal mass transport.

- This gives

$$C = \frac{C_{\text{bulk}}}{\left(\frac{1}{R} - \frac{1}{r_c}\right)}\left(\frac{1}{r} - \frac{1}{r_c}\right)$$

- Moles per second of B reacting equals the moles per second of B entering solid

$$= D_e \left(\frac{dC}{dr}\right)_{r=R} 4\pi R^2$$

  - Equivalently, the moles per second of B reacting equals the moles per second of B reaching the core

$$= D_e \left(\frac{dC}{dr}\right)_{r=r_c} 4\pi r_c^2$$

- Both these expressions lead to the moles per second of B reacting

$$= \frac{-D_e C_{\text{bulk}} 4\pi}{\left(\frac{1}{R} - \frac{1}{r_c}\right)}$$

- Moles per second of solid A reacting

$$= \frac{-D_e C_{\text{bulk}} 4\pi}{\left(\frac{1}{R} - \frac{1}{r_c}\right)} \frac{\nu_A}{\nu_B}$$

- But moles per second of solid A reacting

$$= \frac{-d}{dt}\left(\frac{4}{3}\pi r_c^3 \rho_m\right)$$

  where ρ_m is the molar density of the solid

- Hence

$$-4\pi r_c^2 \rho_m \frac{dr_c}{dt} = \frac{-D_e C_{\text{bulk}} 4\pi}{\left(\frac{1}{R} - \frac{1}{r_c}\right)} \frac{\nu_A}{\nu_B}$$

$$\int_R^{r_c} \left(\frac{1}{R} - \frac{1}{r_c}\right) r_c^2 \, dr_c = \int_0^t \frac{D_e C_{\text{bulk}}}{\rho_m} \frac{\nu_A}{\nu_B} \, dt$$

- Assuming D_e is constant

$$\frac{1}{3R}(r_c^3 - R^3) - \frac{1}{2}(r_c^2 - R^2) = \frac{D_e C_{\text{bulk}}}{\rho_m} \frac{\nu_A}{\nu_B} t$$

$$\frac{r_c^3}{3R} - \frac{r_c^2}{2} + \frac{R^2}{6} = \frac{D_e C_{\text{bulk}}}{\rho_m} \frac{\nu_A}{\nu_B} t$$

$$2\left(\frac{r_c}{R}\right)^3 - 3\left(\frac{r_c}{R}\right)^2 + 1 = \frac{6 D_e C_{\text{bulk}}}{\rho_m R^2} \frac{\nu_A}{\nu_B} t$$

- The time for complete reaction is the time taken to reach $r_c = 0$

$$\tau_c = \frac{\rho_m R^2}{6 D_e C_{\text{bulk}}} \frac{\nu_B}{\nu_A}$$

- Hence

$$2\left(\frac{r_c}{R}\right)^3 - 3\left(\frac{r_c}{R}\right)^2 + 1 = \frac{t}{\tau_c}$$

- The fractional conversion of the reactant is

$$X = 1 - \frac{\frac{4}{3}\pi r_c^3 \rho_m}{\frac{4}{3}\pi R^3 \rho_m} = 1 - \left(\frac{r_c}{R}\right)^3$$

- So

$$\frac{r_c}{R} = (1 - X)^{1/3}$$

- The relationship between conversion and time is

$$2(1 - X) - 3(1 - X)^{2/3} + 1 = \frac{t}{\tau_c}$$

2.2.2 Shrinking core with chemical reaction as the slowest step

- If internal diffusion is rapid, the concentration in the outer zone will be constant and

$$C_{\text{core surface}} \approx C_{\text{particle surface}} \approx C_{\text{bulk}}$$

- The overall kinetics are controlled by chemical reaction at the inner core.
  - We therefore get the same kinetic behaviour as for the shrinking particle model discussed in Section 2.1.2, namely

$$\tau_c = \frac{R\rho_m}{k_{SA} C_{B,\text{bulk}}} \frac{1}{\nu_A}$$

$$X = 1 - \left(1 - \frac{t}{\tau_c}\right)^3$$

2.2.3 Shrinking core with external mass transport as the slowest step

- In this case chemical reaction is rapid so $C_{\text{core surface}} \approx 0$, and internal diffusion is rapid so $C_{\text{particle surface}} \approx 0$.
  - This situation is rare because internal diffusion is generally slower than external transport. However, it could occur for very small particles.
- Moles per second of gas B travelling to surface

$$\approx 4\pi R^2 k_g \left(C_{B,\text{bulk}} - 0\right)$$

- Analysis of this case (see problem sheet) gives

$$X = \frac{t}{\tau_c}$$

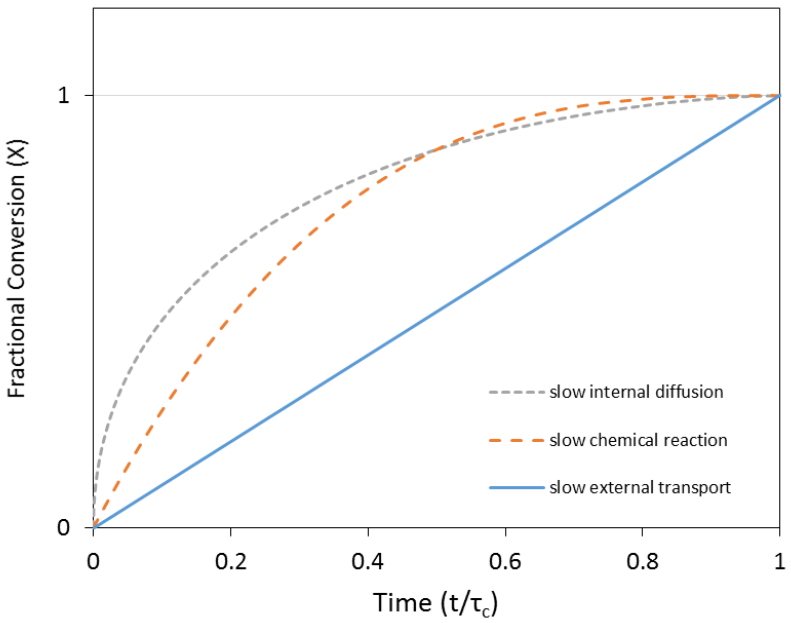


Figure: Fractional conversion for the limiting cases of the shrinking core model

2.2.4 Other factors to consider

- It is straightforward to derive results for particles of other shapes (flat plate, cylinder) in the same way we have done for spherical particles. Results are given for the shrinking core model on page 580 of Levenspiel.
- The reaction may be exothermic (or endothermic) in which case the temperature will increase (or decrease). We may need to do a simultaneous energy balance.
  - There may be a temperature profile outside the particle
  - There may be a temperature profile within the particle
- Temperature dependences:
  - External mass transport k_g and effective diffusion coefficient D_e have a weak temperature dependence
  - Rate constant k_{SA} has a strong temperature dependence.
- The starting particle may be porous. In that case, reaction can take place within the particle, rather than just at the external surface (or external core surface).
- The particle may change size significantly and/or the outer region may flake off.
- Expect $D_e < D$
- All these complications happen when coal is burnt – but that is beyond the scope of this course.

2.2.5. Example of the effect of temperature – evaporation of a droplet

- Consider a spherical liquid droplet of substance A evaporating in a gas.
- Ignoring the effect of varying temperature, this is a case of the shrinking particle model with external mass transfer as the slowest step. Note the concentration gradient is the other way around here – vapour is moving **away** from the surface.
- From Section 2.1.1:

Moles of vapour evaporating from the droplet per second

$$= 4\pi r^2 k_{\mathrm{g}}\left(C_{\mathrm{A,surface}} - C_{\mathrm{A,bulk}}\right)$$

If the gas is dry,

$$C_{\mathrm{A,bulk}} = 0$$

If we have vapour-liquid equilibrium at the droplet's surface,

$$C_{\mathrm{A,surface}} = \frac{P^{\mathrm{sat}}}{RT}$$

Therefore the rate of shrinkage of the droplet is given by:

$$4\pi r^2 k_{\mathrm{g}} \frac{P^{\mathrm{sat}}}{RT} = \frac{-d}{dt}\left(\frac{4}{3}\pi r^3 \rho_{\mathrm{m}}\right)$$

where ρ_{m} is the liquid molar density.

$$\frac{dr}{dt} = -\frac{k_{\mathrm{g}} P^{\mathrm{sat}}}{RT\rho_{\mathrm{m}}}$$

- However, evaporation requires latent heat, so this is equivalent to an endothermic reaction. Heat must be supplied to the droplet and so a thermal gradient is required as well as the concentration gradient.
- Rate of heat transfer **to** the droplet per second

$$= 4\pi r^2 k_{\mathrm{g}} \frac{P^{\mathrm{sat}}}{RT} \Delta H^{\mathrm{vap}} = 4\pi r^2 h(T_0 - T)$$

where ΔH^{vap} is the enthalpy of evaporation of the liquid; h is the heat transfer coefficient to the droplet; T_0 is the temperature in the bulk of the gas; T is the temperature at the surface of the droplet.

- For a gas,

$$Sh = \frac{k_g d}{D} \approx Nu = \frac{hd}{\lambda}$$

where d is the droplet diameter; D is the diffusion coefficient of A in the surrounding gas; and λ is the thermal conductivity of the surrounding gas.

  - This is exactly correct in a stagnant medium when $Sh = Nu = 2$ and correct to within 5% over the full range of Reynolds numbers. It is a consequence of the Reynolds analogy in the turbulent regime and in the laminar regime a result of $Pr \approx Sc$.

- So

$$D\frac{P^{\text{sat}}}{RT}\Delta H^{\text{vap}} = \lambda(T_0 - T)$$

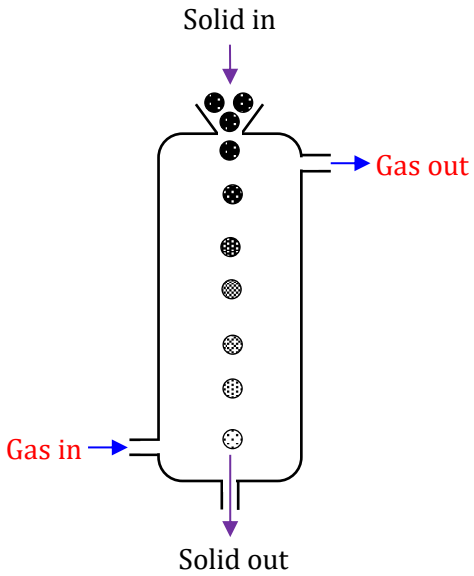
  - ΔH^{vap}, D and k are approximately independent of temperature, but P^{sat} is a strong function of temperature via the Clausius-Clapeyron equation:

$$\ln\left(\frac{P^{\text{sat}}(T)}{P^{\text{sat}}(T_0)}\right) = \frac{\Delta H^{\text{vap}}}{R}\left(\frac{1}{T_0} - \frac{1}{T}\right)$$

- Thus for an evaporating droplet the temperature at the surface is lower than that of the bulk gas and so the saturated pressure of the evaporating liquid is also lower than it would be at the bulk gas temperature. The droplet therefore evaporates slower than it would do at uniform temperature.
- Note that there is no size (r) dependence of T and therefore P^{sat} and so once T has been determined, the problem can be solved in the normal way for an external mass transfer controlled shrinking particle.

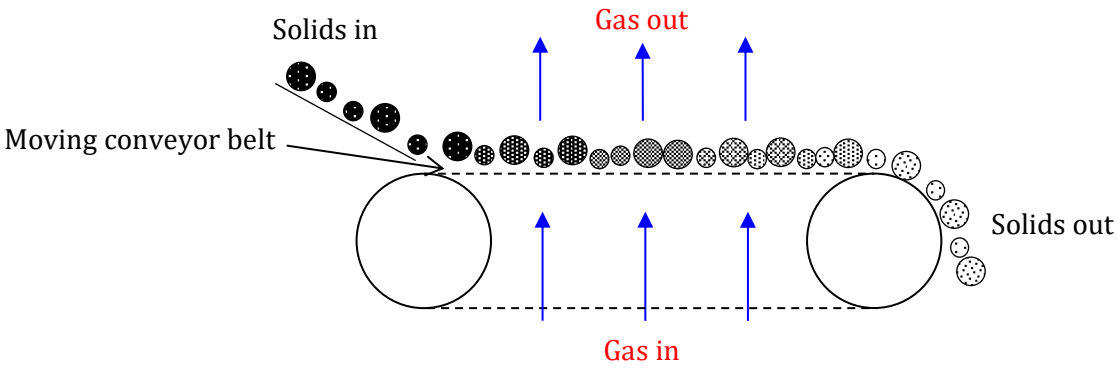
2.3 Types of reactor

- There are a variety of designs for reactors in which the solid is a reactant. Some examples are shown below.
- Falling particle continuous reactor (for short residence times).

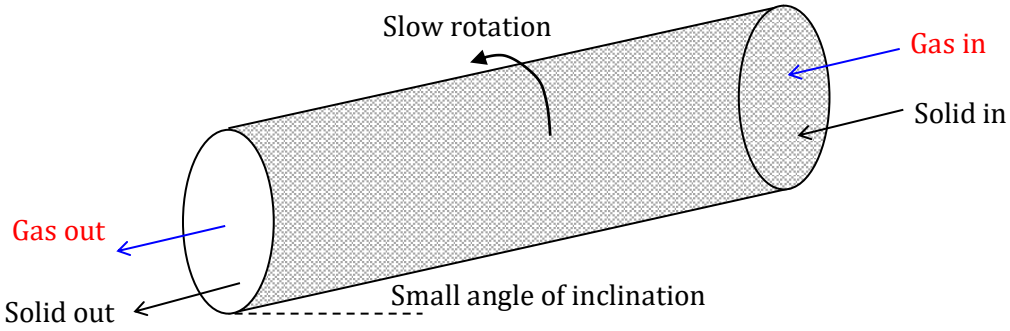


  - In many cases, the solid is pulverised to very small particles and fed to the reactor pneumatically (rather than through gravity).

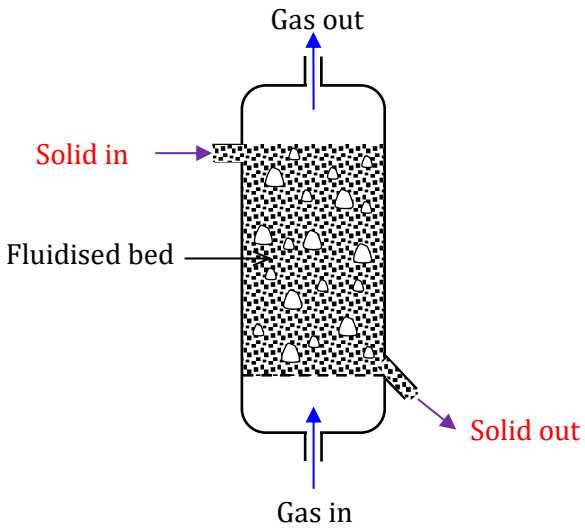
- Moving feed continuous reactor (for medium residence times)



- Rotary kiln continuous reactor (for long residence times)



- Fluidised bed continuous reactor



- Fixed bed semi-batch reactor

